

Exam Computational Social Science with Images and Audio

Multiple choice instructions

For each multiple-choice question, one or two options can be correct. Write the solution sequence for a given question in a text editor such as Word. That is, if you think that the correct answers are true, true, false, true $\rightarrow$ you will write TTFT for that question. In each multiple-choice question, you can get a maximum of 1 point. To avoid random guessing, we will subtract 0.25 points for wrong answers but only within questions (you cannot get less than 0 points for a given question). If you are not sure, you can also give no answer for a given item. For example, if your answer is true, true, don't know, true, you can write TT?T. We will not subtract any points for "don't know/?" answers.

Open questions instructions

For the open questions, the maximum points are indicated for each question.

General remarks

In total, you can get 60 points in this exam, and the exam lasts 60 minutes. That is, to plan your time, you can budget one minute per point. Good luck.

1 Multiple-choice questions

1. How is a single-channel image represented in its simplest form?
 - a) As a three-dimensional array
 - b) As a two-dimensional function mapping coordinates to intensity values
 - c) As a vector of pixel values
 - d) As a set of RGB values
2. Suppose that, by accident, you load an RGB image as BGR in Python (that is, you mistakenly interpret the pixel values for the red channel as the pixel values for the blue channel and vice versa). If you do not correct your mistake, which of the following subsequent outputs will be influenced by this mistake?
 - a) Haar-like features
 - b) Histogram of oriented gradients (HOG)
 - c) Local entropy
 - d) Color histogram
3. Consider the following statements about the impact of images in political and social contexts. Which statements are true?
 - a) Images can influence perceptions of political candidates
 - b) The semantic content of an image often has a low impact on a subject's emotional response to that image
 - c) Images are typically seen as less effective than written content in capturing attention

- d) Casas & Webb Williams', in their work on Black Lives Matter (as discussed in a student presentation), find that angry images generate the most retweets
4. In the context of image analysis, what does entropy measure?
- The colorfulness of the image
 - The randomness or complexity of the texture in the image
 - The number of pixels in the image
 - The orientation of objects in the image
5. In a CNN, what do different layers learn?
- Low-level features like lines and edges
 - Complex abstract features by combining lower-level features
 - Interpret extracted features for the classification task
 - Direct mapping from input pixels to output categories
6. Consider matrix M and filter W :
- The matrix M (3x3) is given by:
- $$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$
- The weight matrix W (2x2) is given by:
- $$W = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
- Convolve M with W , that is: $M * W$. Assume stride 1 and no padding.
- The dimension of the output is 3×2
 - The value at position (1, 1) of the resulting output matrix is 6
 - The value at position (1, 2) of the resulting output matrix is 12
 - If, instead of M , we had an RGB representation of a 3×3 image, we would typically use a separate filter for each color channel
7. What does each filter in a convolutional layer of a CNN produce?
- A unique feature map
 - A copy of the input image
 - A binary classification result
 - A histogram of oriented gradients
8. Which of the following are hyperparameters in CNN?
- Epochs/iterations
 - Learning rate

- c) Dropout rate
 - d) Batch size
9. How does the dropout rate in a CNN model affect its performance?
- a) Increases the model's training accuracy
 - b) Helps in avoiding overfitting
 - c) Speeds up the training process
 - d) Determines the size of the feature maps
10. What role does the activation function play in CNN?
- a) It applies nonlinear transformations to the data
 - b) It helps in pooling the features
 - c) It is used for normalizing the input data
 - d) It defines the structure of the convolutional filters
11. Which of the following statements about human hearing are true?
- a) Humans typically hear frequencies above 20 kHz
 - b) Humans typically hear across 15 octaves
 - c) It is easier for humans to locate low-frequency sounds compared to high-frequency sounds
 - d) While younger humans can hear up to higher frequencies than older people, older people can hear down to lower frequencies than younger humans
12. Which of the following statements about sound waves are true? Sound waves...
- a) ... are a type of electromagnetic radiation
 - b) ... are created by vibrations traveling through a medium like air
 - c) ... can travel through a vacuum
 - d) ... are an electrical signal processed by the brain
13. What does Hertz (Hz) measure in the context of sound waves?
- a) The speed of the sound wave
 - b) The number of cycles per second of a waveform
 - c) The amplitude of the sound
 - d) The duration of a single sound wave
14. Which of the following statements about octaves are true?
- a) An octave is a type of sound wave
 - b) If we double the frequency of a sound wave, we move up an octave
 - c) If we double the frequency of a sound wave, it means that we change the sampling rate
 - d) If we quadruple (i.e., $\times 4$) the frequency of a sound wave, we move up four octaves

15. Which of the following definition(s) accurately describe the sampling rate in digital audio processing?
- a) The speed at which an audio file is played back
 - b) The number of samples per second from a continuous signal to make a digital signal
 - c) The highest frequency that can be accurately represented
 - d) The duration of one cycle of a sound wave
16. How is wavelength related to sound?
- a) It is unrelated to sound and only applies to light
 - b) It refers to the amplitude of the sound wave
 - c) It is the physical distance between identical points in consecutive cycles of a sound wave
 - d) It is the time it takes for a sound wave to complete one cycle
17. What do Mel-Frequency Cepstral Coefficients (MFCCs) represent in audio processing?
- a) The amplitude of sound waves
 - b) The power spectrum aligned with human hearing
 - c) The raw frequency content of the audio
 - d) The duration of the audio signal
18. How are MFCCs typically obtained from audio signals?
- a) By directly sampling the audio waveform
 - b) By dividing the audio signal into overlapping frames and processing each frame
 - c) By applying a Fourier Transform to the entire audio clip
 - d) By averaging the frequencies in the audio signal
19. What does a larger spectral centroid value indicate about a sound?
- a) It is more noise-like
 - b) It has more low frequencies
 - c) It has more high frequencies
 - d) It is more tonally pure
20. What is the purpose of the Fourier Transform in audio analysis?
- a) To extract MFCCs from the audio
 - b) To decompose a signal into its constituent frequencies
 - c) To increase the amplitude of the audio signal
 - d) To convert the audio signal into a digital format
21. What would the Discrete Fourier Transform (DFT) of a single-frequency audio wave show?
- a) A broad range of frequencies
 - b) A peak at the frequency of the sine wave

- c) No significant information
 - d) A flat spectral response
22. How are spectrograms used in audio analysis with CNN?
- a) To reduce the dimensionality of the audio data
 - b) To transform audio data into time-frequency representations
 - c) To encode audio data into binary format
 - d) To compress audio data for efficient storage
23. In the context of RNN, what does the hidden state h_t represent?
- a) The final output of the network
 - b) The memory of the network at time t
 - c) The input data at time t
 - d) The error of the network at time t

2 Open Questions

1. Dietrich used motion detection to measure social polarization in the U.S. House of Representatives. In light of Dietrich's results, should the U.S. introduce a random seat order in the House of Representatives? Why or why not? Discuss. *(6 points)*
2. Image and audio processing tools can be used to modify existing images or audio files. Explain why this can be useful for research experiments and give at least one example from a student presentation in our class. *(6 points)*
3. As a researcher, you observe that a certain demographic group is strongly underrepresented in images of children's books (e.g., the share of images in books showing that group is much lower than the group's share in the population). You also observe that the members of that group have, on average, a significantly lower income than the average income in the population. Explain how these two observations could be related. *(6 points)*
4. In class, we discussed some techniques that allow for identifying "fake" or wrongly contextualized images or videos on social media. Describe two of these techniques. *(6 points)*
5. Suppose you want to solve a classification problem. For such a problem, can you think of a case where you would use the raw pixel values as an input to your machine learning model? Explain. *(5 points)*
6. You have an audio recording that lasts for 3 seconds. Assume it's sampled at the standard rate commonly used for CDs. When examining the underlying vector of this recording (for instance, in Python), what will be its dimension? *(3 points)*
7. Consider these two statements:
 - Often, social science concerns itself with studying social constructs (e.g., partisanship).
 - When manually labeling data, it is important to document the agreement across annotators.

Can you see a relationship between these two statements? Explain why or why not. *(5 points)*